

DOES ANYTHING BEAT A FACTOR MODEL?
COMPARING PREDICTIVE ACCURACY IN LARGE PANELS
OF MACROECONOMIC TIME SERIES

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46th ISF — Montreal, June 2026

Introduction

- **Factor models:** *predominant framework for research on macroeconomic forecasting using many predictors* [Stock–Watson (2012)]

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 - Stock and Watson (2002): diffusion indices cut MSE by roughly one third relative to AR benchmarks. Confirmed across many settings. [Kim–Swanson (2014, 2018), Stock–Watson (2012), ...]
 - Regularisation and model averaging methods also enter the comparison — gains are most consistent when combined with factors, not as a substitute. [Kim–Swanson (2014), Kotchoni et al. (2019), ...]
 - **Factor models are a good benchmark, but evidence varies across series, specifications, and evaluation periods.** [Eickmeier–Ziegler (2008)]

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 - Individual SPA test [White, 2000; Hansen, 2005]: multiple alternatives, single series.
 - Individual DM/SPA tests on cross-sectional average.

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How to assess the superior performance of a benchmark over all alternatives across an entire panel of time series?

① DSPA test — *Dominated Superior Predictive Ability*

- Extends the standard SPA test to N series and M competing methods.
- **Benchmark: factor model** — ARDI [Stock–Watson, 2002] — a natural reference given its dominant role in macro forecasting.

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② Empirical application — FRED-MD

- 126 macro series; 27 competing methods in 7 classes.
- Targets: policy relevant variables (PRV); FRED categories & full database.
- Horizons $\tau = 1, 3, 6, 12$; direct forecasts.
- Rolling estimation window of 120 months; evaluation sample 1970:01–2025:09.

Methodology

- N series, $M + 1$ models; T out-of-sample periods.
- **Model $k = 0$:** factor model (benchmark).
- **Models $k = 1, \dots, M$:** competing alternatives.
- **Loss differential** for model k , series i , period t :

$$d_{k,i,t} = L_{0,i,t} - L_{k,i,t}$$

where $L_{k,i,t}$ is the squared-error loss of model k for series i at time t .

- **Expected gain:** $\mu_{k,i} = \mathbb{E}[d_{k,i,t}]$.
 - $\mu_{k,i} > 0 \Leftrightarrow$ model k beats the benchmark for series i .
- **Primitives:** $d_{k,i,t}$ treated directly — $\mathbf{d}_t$ strictly stationary and α -mixing (serial dependence); cross-series dependence unrestricted, preserved by the bootstrap. Non-growing estimation window [Giacomini–White, 2006] (rules out recursive/expanding schemes; covers nested comparisons).
- **Sample analog:** $\bar{d}_{k,i} = T^{-1} \sum_{t=1}^T d_{k,i,t}$, with HAC standard error $\hat{\omega}_{k,i}$.

DSPA: null hypothesis and test statistic

Hypotheses

The factor model dominates all alternatives across all N series:

$$H_0 : \max_{k=1,\dots,M} \max_{i=1,\dots,N} \mu_{k,i} \leq 0$$

Some alternative outperforms the factor model for at least one series:

$$H_A : \max_{k=1,\dots,M} \max_{i=1,\dots,N} \mu_{k,i} > 0$$

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- **Test statistic** (double maximum over methods and series):

$$\mathcal{T}_{M,N} = \max \left\{ 0, \max_{k,i} \frac{\sqrt{T} \bar{d}_{k,i}}{\hat{\omega}_{k,i}} \right\}$$

- At $N = 1$: reduces to the SPA test of Hansen (2005). [[panel generalisation](#)]
- **Critical values**: stationary bootstrap [Politis–Romano, 1994] with data-dependent recentering under the least favorable configuration [Hansen, 2005]. [▶ algorithm](#)
- **Iterative p-values** can be computed [Sullivan–Timmermann–White, 1999; White, 2000].

Null hypothesis (PDSPA- δ)

The benchmark dominates all alternatives for at least a δ -fraction of the panel:

$$H_0^{P\delta} : \quad Q_\delta\left(\max_{k=1,\dots,M} \mu_{k,i}; \ i = 1, \dots, N\right) \leq 0$$

where $Q_\delta(\cdot)$ denotes the $\lceil \delta N \rceil$ -th order statistic from below.

Null hypothesis (PDSPA- δ)

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where $Q_\delta(\cdot)$ denotes the $\lceil \delta N \rceil$ -th order statistic from below.

- $\delta = 1$: reduces to DSPA (dominance required for every series).
- $\delta = 1/N$: reduces to PDSPA (dominance required for at least one series).
- Intermediate δ : partial dominance — useful when strict uniformity is too strong.
- Same stationary bootstrap scheme; same data-snooping correction.

Simulation Evidence

1 Baseline — as in Hansen (2005)

- Losses follow a location-scale model:

$$L_{k,i,t} = \frac{\lambda_k}{\sqrt{T}} + \sigma_k \varepsilon_{k,i,t}, \quad \varepsilon_{k,i,t} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad \sigma_k^2 = \frac{1}{2} \exp(\arctan(\lambda_k))$$

- $k = 0$ (benchmark): $\lambda_0 = 0$. $k = 1$ (target): $\lambda_1 \in \{0, -1, \dots, -5\}$.
- $k = 2, \dots, M$: nuisance ($\lambda_k = 0$) — noise, not signal; stress-tests size control.
- H_0 holds iff $\lambda_1 \geq 0$; size case is $\lambda_1 = 0$.
- Grid: $M \in \{1, 10, 30, 50\}$, $N \in \{1, 10, 100\}$.

2 Cross-series — equicorrelated losses

- Replace $\varepsilon_{k,i,t}$ with a common-factor structure:

$$\varepsilon_{k,i,t} = \sqrt{1 - \rho_{CS}^N} \eta_{k,i,t} + \sqrt{\rho_{CS}^N} \zeta_{k,t}, \quad \eta_{k,i,t}, \zeta_{k,t} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$$

- $M = 30$ (calibrated to empirical); $\rho_{CS}^N \in \{0, 0.3, 0.5, 0.7, 0.9\}$.
- Common: $T = 1,000$; $B = 500$ bootstrap samples; MC = 2,000 replications.

- DSPA is properly sized across all parameter configurations. [▶ size table](#)
- **Panel (a)** — $M = 30$, vary N and ρ_{CS} :
 - Power increases with N : a larger panel yields stronger evidence against the null.
 - Cross-series correlation attenuates the N -power gain: highly correlated series carry near-redundant information — but the gain never vanishes.
- **Panel (b)** — $\rho_{CS} = 0$, vary M and N :
 - More methods ($\uparrow M$) reduce power, but a large N compensates.

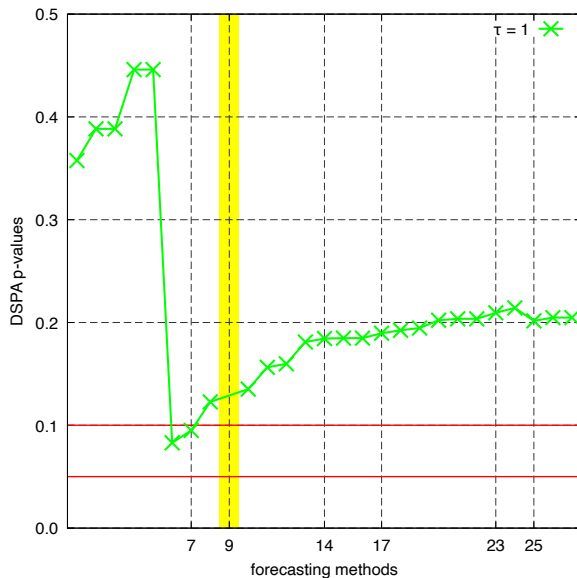
Empirical Application

- **Dataset:** FRED-MD [McCracken–Ng, 2016], 126 monthly macro series; vintage 2025:09.
 - 8 categories: Output/Income, Labour Market, Housing, Consumption/Orders, Money/Credit, Interest/Exchange Rates, Prices, Stock Market.
- **Competing methods:** 27 in 7 groups [▶ details](#)
 - Groups I–II: univariate (AR, RW) and penalised regression (Lasso, Elastic Net).
 - Groups III–V: diffusion indices and complete-subset/ridge-CSR variants.
 - Group VI: Random Forest.
 - Group VII: forecast combinations.
- **Benchmark:** factor model (ARDI, Group III).
- **Evaluation:** MSE; rolling window ($R = 120$ months); $\tau = 1, 3, 6, 12$.
- **Two panels:** policy-relevant variables ($N = 10$) and full FRED-MD ($N = 126$).

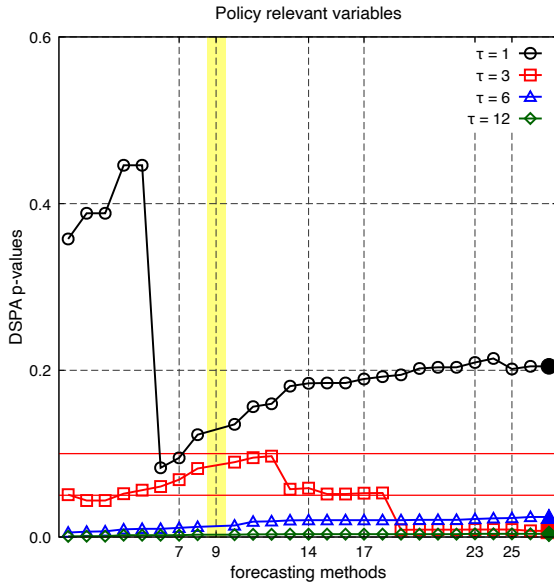
PRV: relative MSE and p-values ($\tau = 1$) ▶ $\tau = 12$

	IP	RPI	UR	PAY	HST	M2	T10	CPI	PPI	SNP	All (avg. rel. MSE, rank)
(1) CONST	0.974	0.496	0.614	0.140	>10	0.643	1.077	1.024	0.998	1.005	1.810 (21)
(2) RW	0.983	0.508	0.607	0.144	>10	0.637	1.069	1.015	0.990	1.027	>10 (27)
(3) RWD	0.974	0.496	0.614	0.140	>10	0.643	1.077	1.024	0.998	1.005	1.810 (22)
(4) ARDIRECT	1.839	0.507	5.082	>10	0.701	0.705	0.985	0.928	0.943	0.970	2.396 (25)
(5) ARI	1.839	0.507	5.082	>10	0.701	0.705	0.985	0.928	0.943	0.970	2.396 (26)
(6) ARMAI	1.440	0.480	5.497	9.724	0.671	0.641	0.938**	0.820**	0.804**	0.994	2.201 (24)
(7) Lasso	1.487	0.580	1.436	2.209	0.599	1.473	1.120	1.123	1.849	1.163	1.304 (13)
(8) ElNet	1.484	0.584	1.608	2.120	0.600	1.468	1.126	1.116*	1.844	1.164	1.312 (14)
(9) ARDI	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000 (6)
(10) ARDI-soft	0.853	0.767	0.708	0.643	0.732	1.177	1.143	0.910*	0.916	1.068	0.892 (3)
(11) ARDI-hard, 1.28	2.404	1.067	1.348	0.864	0.936	1.078	1.027	0.931	0.964	1.025	1.164 (10)
(12) ARDI-hard, 1.65	2.313	1.083*	1.478	0.833	1.004	1.020	1.115	0.884	0.975	1.047*	1.175 (11)
(13) ARDI-tstat, 1.96	1.378	0.772	1.085	1.002	1.028	0.660	1.016	0.917	0.933	1.015*	0.980 (5)
(14) CSR, 1	1.722	0.433	5.071	8.684	0.711*	0.680	0.981	0.966	0.980	0.957	2.119 (23)
(15) CSR, 10	2.382	0.751	4.367	3.556	0.679	0.948	0.952	0.936	0.959	0.968*	1.650 (20)
(16) CSR, 20	2.171	0.738	3.371	1.907	0.652*	1.033	0.987	0.928	0.940	1.027	1.375 (16)
(17) R-CSR, 10	2.315	0.739	4.262	3.533	0.681	0.933	0.953	0.935	0.958	0.968	1.628 (19)
(18) R-CSR, 20	2.198	0.755	3.386	1.928	0.656	1.035	0.984	0.930	0.943	1.029	1.384 (17)
(19) T-CSR-hard, 1.65, 10	2.256	0.731	3.653	1.858	0.634*	1.895	1.049	0.891	0.982	1.094	1.504 (18)
(20) T-CSR-hard, 1.65, 20	1.839	0.639*	2.510	0.958*	0.672*	2.295	1.123	0.928	1.008	1.210	1.318 (15)
(21) T-CSR-soft, 10	1.229	0.541*	2.216	1.344	0.552*	1.564	1.029	0.904	0.910	1.079	1.137 (9)
(22) T-CSR-soft, 20	1.291	0.543	1.380	0.759	0.617*	1.579	1.455	1.379	1.399	1.400	1.180 (12)
(23) RFOREST	0.911	0.482	0.610	0.156	1.040	0.625	1.046	0.990	0.945	1.017	0.782 (2)
(24) RFORESTX	0.865	0.544	0.691	0.140	0.899	0.652	1.061	0.984	0.965	1.010	0.781 (1)
(25) Average	1.375	0.530*	1.969	1.792	0.618*	0.813	0.939**	0.790**	0.803**	0.966**	1.059 (8)
(26) Median	1.366	0.552*	1.749	0.970*	0.598*	0.885	0.948*	0.829**	0.845**	0.971**	0.971 (4)
(27) T-Average	1.399	0.532*	1.973	1.676	0.588*	0.802	0.939**	0.792**	0.804**	0.965**	1.047 (7)
SPA/DSPA p-values	0.48	0.19	0.31	0.18	0.20	0.48	0.17	0.05	0.06	0.13	0.20

Policy-relevant variables ($N = 10$): short run ► series list

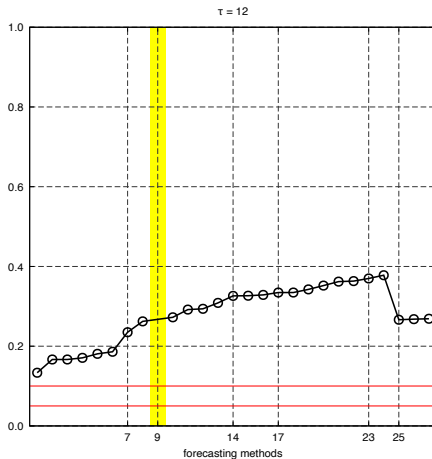
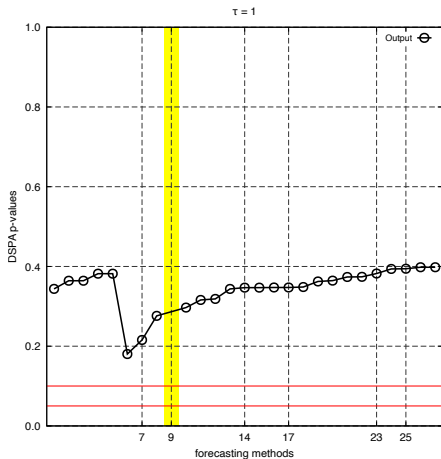


Policy-relevant variables ($N = 10$): all horizons

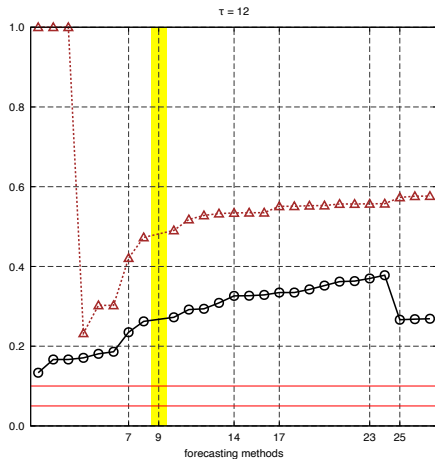
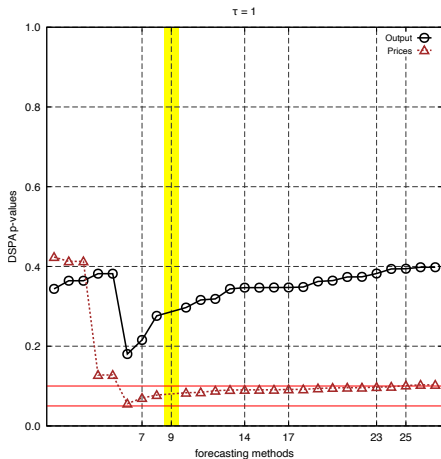
[series list](#)

- Each curve traces the DSPA p -value as methods are added left to right; each dashed vertical line separates a model group.
- **Horizon gradient:**
 - $\tau = 1$: path dips to 0.08 at ARMA1 but rebounds to $p = 0.20$ once all methods are included — data-snooping correction prevents rejection.
 - $\tau = 3$: RW already pushes p below 5%; regularised CSR (method 19) delivers the decisive drop to $p = 0.007$.
 - $\tau = 6, 12$: rejection is immediate.
- **Takeaway:** at longer horizons, even the simplest models beat the factor model uniformly across the PRV panel.

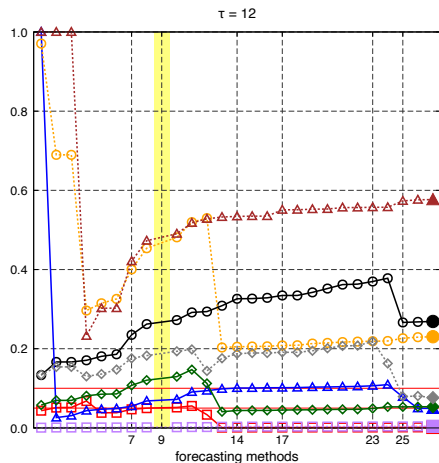
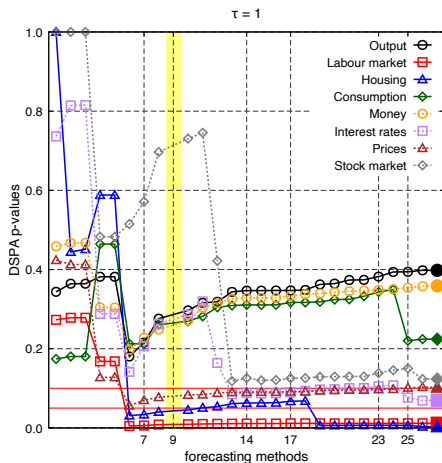
Category heterogeneity: Output & Income



Category heterogeneity: adding Prices

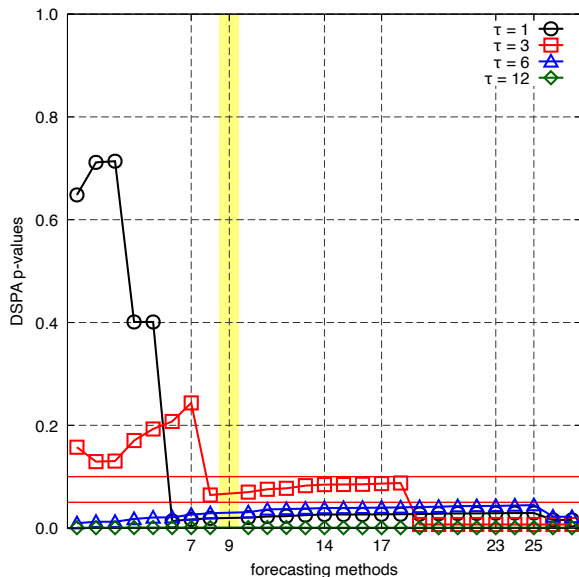


Category heterogeneity: all categories



- **Factor model never dominated** (non-rejection at both horizons):
 - **Output & Income** and **Money & Credit** — precisely where diffusion indices were designed to work best [Stock–Watson, 2002].
- **Factor model consistently dominated** (rejection at both horizons):
 - **Labour Market**: high autocorrelation in unemployment and payrolls; many methods exploit this well.
 - **Interest & Exchange Rates**: domain-specific structure (yield curve, near-random walk at short horizons) that generic factors do not fully capture.
- **Horizon-dependent results**:
 - **Prices**: factor model's advantage grows at long horizons — inflation dynamics become diffuse, favouring information aggregation.
 - **Housing, Consumption, Stock Market**: borderline or mixed.
- **Takeaway**: rejection of factor-model dominance is not arbitrary — it reflects genuine, theoretically interpretable heterogeneity across categories.

Full FRED-MD panel ($N = 126$)



- With $N = 126$ series, **all four horizons reject**:

- $\tau = 1$: $p = 0.030$; $\tau = 3$: $p = 0.007$; $\tau = 6$: $p = 0.046$; $\tau = 12$: $p = 0.001$.

- **Panel power gain**: the 1-month horizon that could not be rejected on 10 series is rejected on 126; labour-market and housing variables already favour simpler alternatives at short leads.

- **Drivers vary by horizon**:

- $\tau = 1$: a univariate model (ARMAI) is already sufficient to drive rejection.
 - $\tau = 3$: penalised regression (EINet) brings p below 10%; regularised CSR (method 19) delivers the decisive drop to $p = 0.007$.
 - $\tau = 6, 12$: rejection is immediate.

- **Robust to**:

- alternative block lengths ($q = 2, 5, 10, 20$);
 - shorter out-of-sample windows (50%–90% of baseline).

Conclusion

1 Methodology

- DSPA extends SPA to a panel of N series: tests uniform outperformance across all series simultaneously.
- PDSPA relaxes to a δ -fraction: a flexible intermediate test.
- Properly sized; power grows with N even under cross-series correlation.

2 Empirical findings

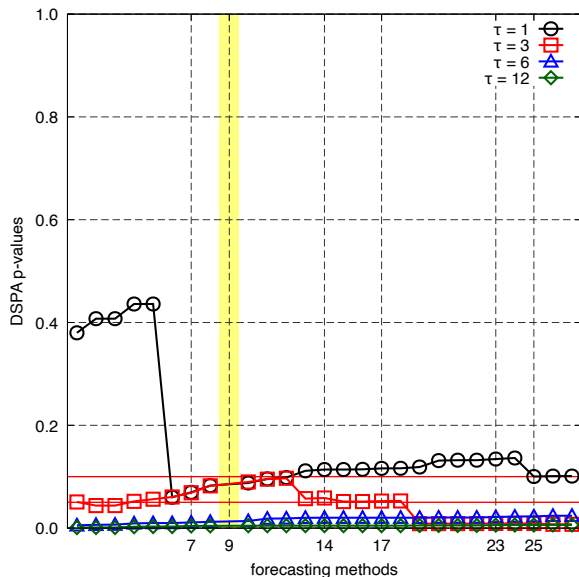
- PRV ($N = 10$): factor model holds at $\tau = 1$; dominance rejected from $\tau = 3$ onward — at longer horizons even simple models suffice.
- **Category heterogeneity**: Labour Market and Interest/Exchange Rates reject at both horizons; Output & Income and Money & Credit never reject — precisely where factor models were designed to work.
- Full FRED-MD ($N = 126$): **dominance rejected at all four horizons** — panel power gain brings $\tau = 1$ rejection that was absent with $N = 10$.
- Results robust to bootstrap specification and evaluation window.

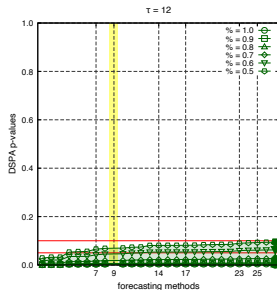
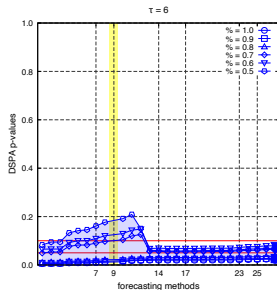
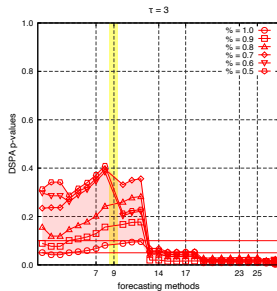
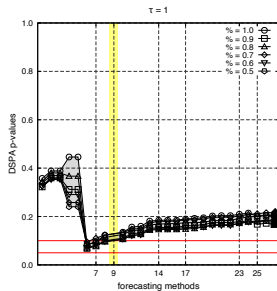
3 Takeaway

- The factor model is a strong benchmark — it cannot be beaten for Output & Income or Money & Credit. Where it is dominated, category heterogeneity tells us exactly why.

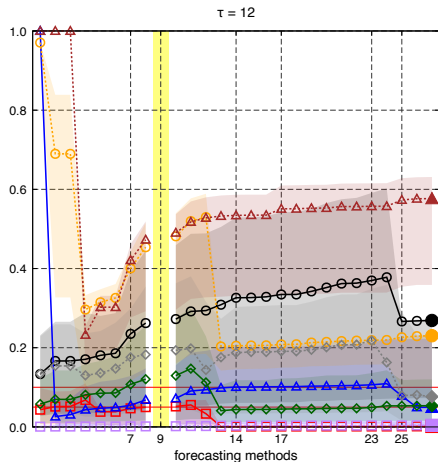
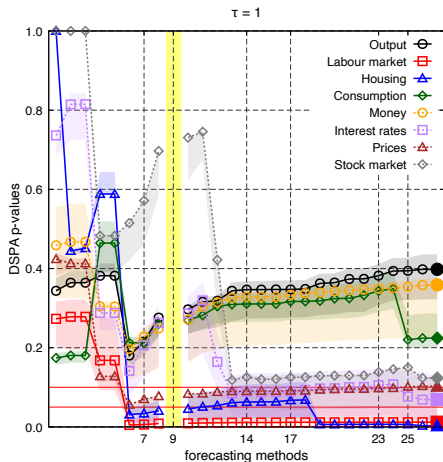
Thank you.

`i.crespo@cunef.edu`

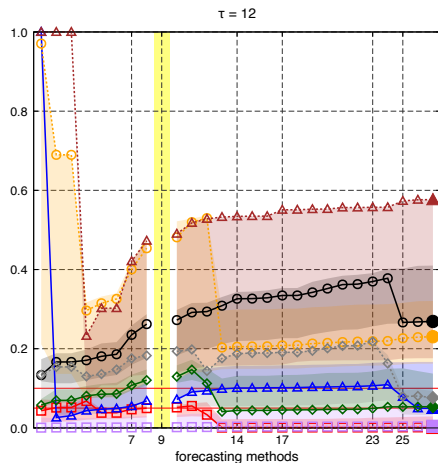
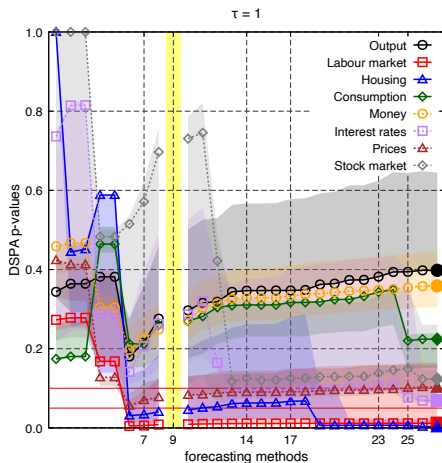




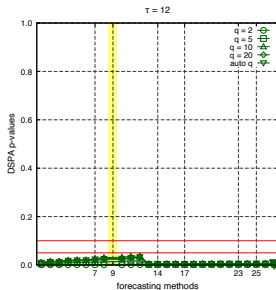
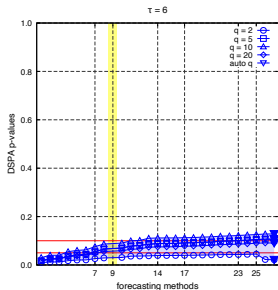
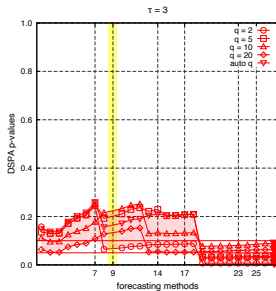
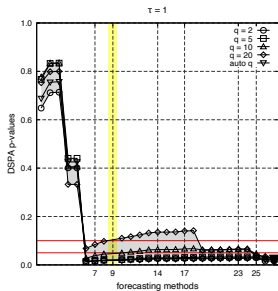
Category heterogeneity: robustness to block length (q)

[► OOS robustness](#)[► back](#)

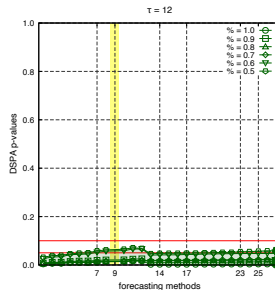
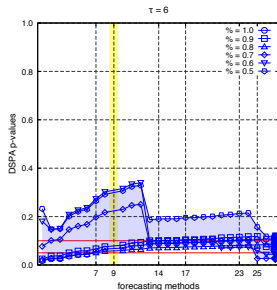
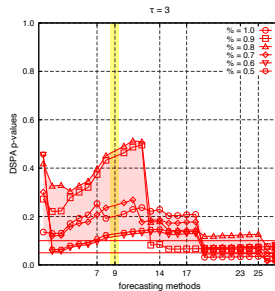
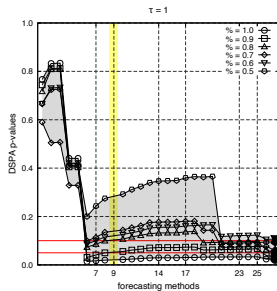
Category heterogeneity: robustness to OOS window

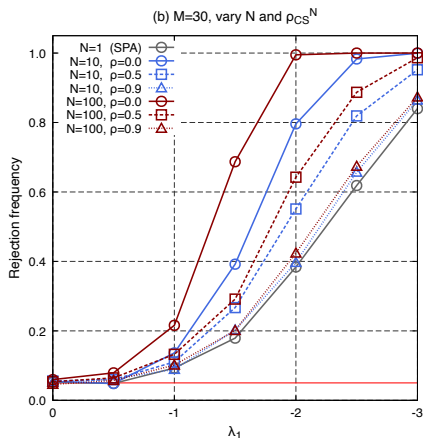
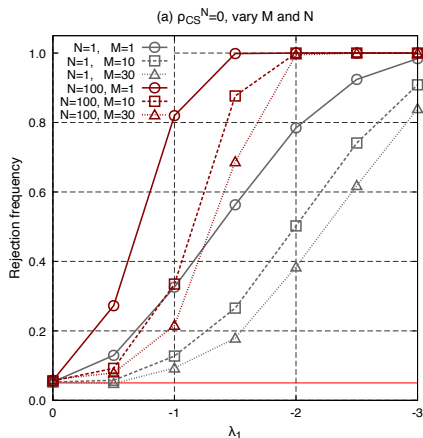
[q robustness](#)[back](#)

Full FRED-MD: robustness to block length (q)

[► OOS robustness](#)[► back](#)

Full FRED-MD: robustness to OOS window

[q robustness](#)[back](#)



- 1 Compute the observed test statistic:

$$\mathcal{T} = \max_{k,i} \frac{\sqrt{T} \bar{d}_{k,i}}{\hat{\omega}_{k,i}}$$

- 2 Draw B stationary bootstrap samples [Politis–Romano, 1994] of $\{\mathbf{d}_t\}$. For each bootstrap sample b :

- Compute the recentered bootstrap mean $\bar{d}_{k,i}^{*b} - \bar{d}_{k,i}^{\text{rec}}$, where [Hansen, 2005]:

$$\bar{d}_{k,i}^{\text{rec}} = \bar{d}_{k,i} \cdot \mathbf{1} \left\{ \frac{\sqrt{T} \bar{d}_{k,i}}{\hat{\omega}_{k,i}} > -\sqrt{2 \log \log T} \right\}$$

The threshold $\sqrt{2 \log \log T}$ separates binding ($\mu_{k,i} = 0$) from slack ($\mu_{k,i} < 0$) constraints; slack alternatives are zeroed out so they do not inflate the critical value.

- Compute the bootstrap statistic $\mathcal{T}^{*b} = \max_{k,i} \frac{\sqrt{T}(\bar{d}_{k,i}^{*b} - \bar{d}_{k,i}^{\text{rec}})}{\hat{\omega}_{k,i}}$.

- 3 The DSPA p -value is:

$$\hat{p} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}(\mathcal{T}^{*b} \geq \mathcal{T})$$

Group	#	Label	Description
I	1–6	CONST, RW, RWD, ARDIRECT, ARI, ARMAI	Univariate methods
II	7–8	Lasso, EInet	Penalised regression
III	9–13	ARDI, ARDI-soft, ARDI-hard ($\times 2$), ARDI-tstat	Diffusion index models
IV	14–16	CSR 1 / 10 / 20	Complete subset regression
V	17–22	R-CSR, T-CSR-hard, T-CSR-soft ($\times 2$)	Regularised CSR variants
VI	23–24	RFOREST, RFORESTX	Random Forest
VII	25–27	Average, Median, T-Average	Forecast combinations

Λ_1	$N = 1$				$N = 10$				$N = 100$			
	$M=1$	$M=10$	$M=30$	$M=50$	$M=1$	$M=10$	$M=30$	$M=50$	$M=1$	$M=10$	$M=30$	$M=50$
0	0.053	0.053	0.052	0.054	0.047	0.056	0.055	0.055	0.057	0.056	0.060	0.060
-1	0.326	0.127	0.093	0.077	0.577	0.210	0.137	0.106	0.820	0.335	0.216	0.157
-2	0.785	0.502	0.384	0.343	0.997	0.918	0.796	0.737	1.000	1.000	0.995	0.980
-3	0.984	0.909	0.840	0.813	1.000	1.000	1.000	0.999	1.000	1.000	1.000	1.000
-4	1.000	0.997	0.989	0.988	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
-5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

$T = 1,000$, $B = 500$, $q = 1$, $MC = 2,000$. $\Lambda_1 = 0$: size; $\Lambda_1 < 0$: power. Columns grouped by N ; within each group $M \in \{1, 10, 30, 50\}$.

Series	Abbr.	Category
Industrial Production	IP	Output and Income
Real Personal Income	RPI	Output and Income
Unemployment Rate	UR	Labour Market
Total Nonfarm Payrolls	PAY	Labour Market
Housing Starts	HST	Housing
M2 Money Stock	M2	Money and Credit
10Y Treasury Rate	T10	Interest/Exchange Rates
Consumer Price Index	CPI	Prices
CPI Less Food	CPIx	Prices
S&P 500 Index	SNP	Stock Market

PRV: relative MSE and p-values ($\tau = 12$) ► back

	IP	RPI	UR	PAY	HST	M2	T10	CPI	PPI	SNP	All (avg. rel. MSE, rank)
(1) CONST	1.045	0.785**	0.747	0.845	1.614	1.206	0.732***	1.806	1.570	0.711*	1.106 (12)
(2) RW	1.020	1.614	0.656*	1.062	>10	1.186	0.668***	1.767	1.541	0.783	>10 (24)
(3) RWD	1.045	0.785**	0.747	0.845	1.614	1.206	0.732***	1.806	1.570	0.711*	1.106 (13)
(4) ARDIRECT	1.067	0.765**	0.713	0.848	1.009	0.694**	0.748***	0.894	0.781	0.694**	0.821 (1)
(5) ARI	>10	0.780**	>10	>10	0.916	1.265	0.749***	1.240	1.138	0.703*	>10 (26)
(6) ARMAI	>10	0.792*	>10	>10	1.717	1.590	0.758***	2.572	2.220	0.754*	>10 (27)
(7) Lasso	3.602	1.214	1.240	2.326	0.810	2.388	1.933	2.216	2.857	1.438	2.002 (22)
(8) ElNet	3.603	1.214	1.214	2.325	0.833	2.384	1.953	2.217	2.857	1.485	2.009 (23)
(9) ARDI	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000 (7)
(10) ARDI-soft	1.122	1.135	1.393	1.695	0.866	1.036	1.286	1.014	0.975	1.000	1.152 (15)
(11) ARDI-hard, 1.28	1.052	1.072	0.956	1.319	0.911**	1.086	1.643	0.923	0.872	1.377	1.121 (14)
(12) ARDI-hard, 1.65	1.148	1.084**	1.106	1.414*	0.921	0.997	1.709***	1.025	0.946	1.263**	1.161 (17)
(13) ARDI-tstat, 1.96	1.181	0.939*	1.104*	1.202	0.854**	0.907	1.245**	0.822	0.809	1.260**	1.032 (8)
(14) CSR, 1	1.007	0.771	0.690	0.835	0.928**	0.992	0.764	1.346	1.205	0.696	0.923 (4)
(15) CSR, 10	0.896	0.860*	0.684*	0.910	0.783**	0.927	0.920**	1.293	1.159	0.888**	0.932 (5)
(16) CSR, 20	1.181	0.991	0.891	1.290	0.746**	1.416	1.124	1.487	1.341	1.153	1.162 (18)
(17) R-CSR, 10	0.907	0.856	0.688	0.923	0.781	0.942	0.921	1.295	1.161	0.899	0.937 (6)
(18) R-CSR, 20	1.182	0.991	0.879	1.271	0.749	1.443	1.125	1.493	1.347	1.144	1.162 (19)
(19) T-CSR-hard, 1.65, 10	1.437	1.107	0.899	1.275	0.817*	0.895	1.350	1.480	1.188	1.095	1.154 (16)
(20) T-CSR-hard, 1.65, 20	2.313	1.467	1.128	1.561	1.116	0.892	1.584	1.793	1.326	1.261	1.444 (20)
(21) T-CSR-soft, 10	1.307	1.056	0.861	1.589	0.725**	0.790*	1.148	1.277	1.160	0.976	1.089 (11)
(22) T-CSR-soft, 20	2.676	1.267	1.893	2.327	0.874	0.942	1.607	1.569	1.339	1.666	1.616 (21)
(23) RFOREST	1.003	0.900	0.965	1.070	0.743*	1.232	0.938	1.460	1.260	0.844	1.041 (10)
(24) RFORESTX	1.079	0.915	0.844	1.021	0.863	1.114	1.045	1.404	1.236	0.808**	1.033 (9)
(25) Average	3.465	0.851**	>10	>10	0.682**	0.701**	1.037	0.928	0.870	0.853**	>10 (25)
(26) Median	0.965	0.855*	0.805	1.093	0.709**	0.760**	1.014	1.048	0.948	0.885**	0.908 (3)
(27) T-Average	0.950	0.846**	0.787	0.996	0.692**	0.707**	1.024	0.927	0.865	0.847**	0.864 (2)
SPA/DSPA p-values	0.87	0.12	0.25	0.31	0.05	0.07	0.00	0.57	0.43	0.05	0.00